

POWERFACTORY

PowerFactory 2024

Technical Reference

Current Measurement

Stalmea

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

The *Current Measurement Device* (*StaImea*) can be used to measure the current flow at a cubicle of any element, which is connected to a terminal/busbar. The measured current can then be fed into a controller. The reference "Measurement Location" defines the cubicle at which the current is measured. If no measurement location is set, the current measurement element should be placed inside a cubicle, which then defines automatically the measured location.

The direction (sign) of the current flow is depending on the element and its connection point. For loads, motors and passive elements (lines, transformers, etc.) the current flow is always defined in load orientation (in the direction of the active power flow). This means the current is always flowing out of the terminal into the element. For sources, e.g. generators, external network, current and voltage sources, etc... the current is in generator orientation and thus out of the element in the direction to the connected bus.

AC, DC and BI-phase system types can be measured by the *StaImea* element. Also, *Number of phases* option is available in the *StaImea*:

- if *Number of phases* =3, not only three-phase, but also two-phase and single-phase elements can be measured (the non-relevant outputs are then set to 0).
- if *Number of phases* = 1, the measured quantities *a, b, c or n* can be selected by the appropriate *Measured Phase*.

The outputs of the measurement device can be selected to be phase or transformed quantities (symmetrical components for LF and RMS calculations and $\alpha\beta\gamma$ components for the EMT):

- *Phase quantities*, depending on the selection of the *p.u. values* three different options are available:
  - *Equivalent to kA values*, where the p.u. output value corresponds to the value in kA,
  - *Based on current rating of connected element* where the p.u. output is based on current rating of the connected element, and
  - *Based on user-defined current* where the p.u. output is calculated using the $Inom$ input parameter as base value.
- *Transformed quantities*, depending on the selection of the *p.u. values* there are three different options:
  - *Based on 1 MVA and nominal voltage*, where the p.u. output is based on 1 kA,
  - *Based on current rating of connected element* where the p.u. output is based on current rating of the connected element, and
  - *Based on user-defined current* where the p.u. output is calculated using the $Inom$ input parameter as base value.

For the EMT calculation, options are also available to measure RMS quantities. This option can be selected on the EMT page and is supported for phase quantities measurements.

The current magnitudes (i, $i2$, $i0$, ...) are calculated by using the real and imaginary part of the corresponding current as follows:

$$i = \sqrt{ir^2 + ii^2} \tag{1}$$

If an element and its controller models are using an A-stable integration algorithm, the connected measurement device should also use this algorithm, thus the flag '*iAstabint*' on the RMS-simulation page should be selected.

The output signals and transformation equations of the *StaImea* element are documented in the following sections.

2 Load Flow

2.1 Phase quantities outputs

The phase quantities output signals available for the Load Flow calculation are identical to the output signals presented in the RMS-Simulation. Please refer to section 3 for more information.

2.2 Symmetrical components

The symmetrical components output signals available for the Load Flow calculation are identical to the output signals presented in the RMS-Simulation. Please refer to Section 3 for more information.

3 RMS-Simulation

The equations for balanced and unbalanced RMS simulation are described in the following subsections.

3.1 Balanced

From the balanced RMS-Simulation, the real and imaginary part of the positive-sequence current are available. Based on these currents, also additional current quantities can be calculated.

3.1.1 Phase quantities

The available measurement outputs are presented in Table 3.1.

Table 3.1: Outputs, Phase quantities for Balanced RMS-Simulation

Name	Unit	Description
$i1r$	p.u.	Positive sequence current, real part
$i1i$	p.u.	Positive sequence current, imaginary part

- For DC systems, only one signal is available $i1r$ ($i1i = 0$).

3.1.2 Symmetrical components

The available outputs for a measurement at a single terminal are presented in Table 3.2.

Table 3.2: Outputs, Symmetrical components for Balanced RMS-Simulation

Name	Unit	Description
ir	p.u.	Positive sequence current, real part
ii	p.u.	Positive sequence current, imaginary part
i	p.u.	Positive sequence current, magnitude

- For DC systems, only one signal is available ir ($ii = 0$).

3.2 Unbalanced

From the unbalanced RMS-Simulation, the real and imaginary parts currents ($\underline{i}_a$, $\underline{i}_b$ and $\underline{i}_c$) are available to the measurement device. Using these available currents, also additional quantities like the positive-, negative-, and zero-sequence currents can be calculated.

The unbalanced measurement depends on the *No. of Phases* option (parameter $nphase$).

3.2.1 Phase quantities

The available measurement outputs are presented in Table 3.3.

Table 3.3: Outputs, Phase quantities for Unbalanced RMS-Simulation

Name	Unit	Description
ii_A	p.u.	Current, phase a, imaginary part
ir_B	p.u.	Current, phase b, real part
ii_B	p.u.	Current, phase b, imaginary part
ir_C	p.u.	Current, phase c, real part
ii_C	p.u.	Current, phase c, imaginary part

3.2.1.1 Single-phase measurement (*Number of phases*$= 1$)

The available measurement outputs are presented in Table 3.4.

Table 3.4: Outputs, Phase quantities for Unbalanced RMS-Simulation, single-phase measurement

Name	Unit	Description
ir	p.u.	Phase current, real part
ii	p.u.	Phase current, imaginary part

3.2.2 Symmetrical components

In the case of a three-phase system, the Symmetrical components are the real and imaginary parts and magnitude of the positive-, negative- and zero-sequence current that are calculated using the symmetrical components transformation. Similar transformation is being used for the two-phase system (modal components transformation).

The available measurement outputs are presented in Table 3.5.

Table 3.5: Outputs, Symmetrical components for Unbalanced RMS-Simulation

Name	Unit	Description
ir	p.u.	Positive sequence current, real part
ii	p.u.	Positive sequence current, imaginary part
i	p.u.	Positive sequence current, magnitude
$i2r$	p.u.	Negative sequence current, real part
$i2i$	p.u.	Negative sequence current, imaginary part
$i2$	p.u.	Negative sequence current, magnitude
$i0r$	p.u.	Zero sequence current, real part
$i0i$	p.u.	Zero sequence current, imaginary part
$i0$	p.u.	Zero sequence current, magnitude

The output currents for three-phase unbalanced RMS simulation are calculated depending on the phase technology of the element as follows:

- 3-phase system (using symmetrical components transformation)

$$ir + j\,ii = \frac{1}{3} \cdot \left(\underline{i}_a + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) \cdot \underline{i}_b + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) \cdot \underline{i}_c \right) \quad (2)$$

$$i2r + j\,i2i = \frac{1}{3} \cdot \left(\underline{i}_a + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) \cdot \underline{i}_b + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) \cdot \underline{i}_c \right) \quad (3)$$

$$i0r + j\,i0i = \frac{1}{3} \cdot (\underline{i}_a + \underline{i}_b + \underline{i}_c) \quad (4)$$

- BI-phase system ($180°$):

$$ir + j\,ii = \frac{1}{2} \cdot (\underline{i}_a - \underline{i}_b) \quad (5)$$

$$i2r + j\,i2i = 0 \quad (6)$$

$$i0r + j\,i0i = \frac{1}{2} \cdot (\underline{i}_a + \underline{i}_b) \quad (7)$$

- 2-phase system ($120°$):

$$ir + j\,ii = \frac{1}{\sqrt{3}} \cdot (\underline{i}_a - \underline{i}_b) \quad (8)$$

$$i2r + j\,i2i = 0 \quad (9)$$

$$i0r + j\,i0i = \frac{1}{\sqrt{3}} \cdot (\underline{i}_a + \underline{i}_b) \quad (10)$$

- Single-phase (*Number of phases*$= 3$):

  The current is set (transformation is not used) to the complex current $\underline{i}$ available from the RMS-Simulation as follows:

$$ir + j\,ii = \underline{i} \quad (11)$$

- For DC systems, only one signal is available ir ($ii = 0$).

3.2.2.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase measurement (*Number of phases*$= 1$) are presented in Table 3.6.

Table 3.6: Outputs, Symmetrical components for Unbalanced RMS-Simulation, single-phase measurement

Name	Unit	Description
ir	p.u.	Current, real part
ii	p.u.	Current, imaginary part
i	p.u.	Current, magnitude

- Single-phase measurement (*Number of phases*$= 1$):

  The current is set (transformation is not used) to the complex current $\underline{i}$ available from the RMS-Simulation for the selected measured phase (*a, b, c, or n*) as follows:

$$ir + j\,ii = \underline{i} \quad (12)$$

4 EMT-Simulation

From the EMT-Simulation, the currents i_a, i_b and i_c are available to the measurement device. Using these currents, additional quantities like the alpha, beta and gamma components of the current can be calculated.

The output signals available for the EMT-Simulation are described in the following subsections.

4.1 Instantaneous values (phase quantities)

The available measurement outputs are presented in Table 4.1.

Table 4.1: Outputs, Instantaneous values (phase quantities) for EMT-Simulation

Name	Unit	Description
i_A	p.u.	Current, phase a
i_B	p.u.	Current, phase b
i_C	p.u.	Current, phase c

4.1.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase measurement are presented in Table 4.2.

Table 4.2: Outputs, Instantaneous values (phase quantities) for EMT-Simulation, single-phase measurement

Name	Unit	Description
i	p.u.	Phase current

4.2 True RMS amplitude values (phase quantities)

The measurement device measuring true RMS values is a clocked device, for which the following input parameters need to be defined:

- *Number of samples per period* (input parameter $samplesPerPeriod$)
- *Calculate (update) outputs every* number of samples (input parameter $samplesForUpdate$)

The resulting sampling frequency in [Hz] with which the calculation buffer is filled at defined intervals is obtained using the nominal frequency $Fnom$ and the input parameter $samplesPerPeriod$ as:

$$samplingFreq = Fnom \cdot samplesPerPeriod \tag{13}$$

The output of the measurement device is calculated at constant intervals after the defined number of samples for updating. The resulting update cycle (parameter $clockPeriod$) in [s] is:

$$clockPeriod = \frac{samplesForUpdate}{samplesPerPeriod \cdot Fnom} \tag{14}$$

The calculation of the true RMS amplitudes from the instantaneous values x_1,x_2, ...$x_{samplesPerPeriod}$ is based on the following equation:

$$RMS = \sqrt{\frac{x_1^2 + x_2^2 + ... + x_{samplesPerPeriod}^2}{samplesPerPeriod}} \tag{15}$$

When the option *Smooth outputs when updated* is used, the outputs are linearly interpolated (smoothed).

The available outputs for a true RMS value measurement are presented in Table 4.3.

Table 4.3: Digital outputs, True RMS amplitude values (phase quantities) for EMT-Simulation

Name	Unit	Description
i_A	p.u.	RMS current A
i_B	p.u.	RMS current B
i_C	p.u.	RMS current C

4.2.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase instantaneous values measurement at a single terminal are presented in Table 4.4.

Table 4.4: Digital outputs, True RMS amplitude values (phase quantities) for EMT-Simulation, single-phase measurement (*Number of phases*$= 1$)

Name	Unit	Description
u	p.u.	RMS current

4.3 RMS phasor and symmetrical components (phase quantities)

The measurement device measuring RMS phasors is a clocked device which calculates a Discrete Fourier Transform (DFT) for a single frequency. There are four different options for estimating the frequency for which the DFT is calculated:

- *Use nominal frequency*

  The DFT is executed for nominal frequency.

- *Phase angle derivative*

  The frequency supplied to the DFT (fe), is estimated based on the phase angle derivative.

  The calculation of the frequency fe is similar to the calculation in the RMS simulation, with the difference that instead the positive-sequence, the dq voltage values are used. The dq quantities of the voltage are obtained as:

$$ud + j\ uq = (ur + j\ ui) \cdot (\cos(phiref) - j\ (\sin(phiref))) \tag{16}$$

  The following differential equations are being used:

$$\frac{d\ \cos\phi}{dt} = \left(\frac{ud}{\sqrt{ud^2 + uq^2}} - \cos\phi \right) \Big/ T_{fe} \tag{17}$$

$$\frac{d\ \sin\phi}{dt} = \left(\frac{uq}{\sqrt{ud^2 + uq^2}} - \sin\phi \right) \Big/ T_{fe} \tag{18}$$

where T_fe is fixed to $3/Fnom$.

The frequency change (in p.u.) is equal to:

$$dfe = \sqrt{\frac{\left(\frac{d\ \cos\phi}{dt}\right)^2 + \left(\frac{d\ \sin\phi}{dt}\right)^2}{(\cos\phi)^2 + (\sin\phi)^2}} \Big/ (2 \cdot \pi \cdot Fnom) \tag{19}$$

The frequency is then calculated as $fe = 1 + dfe$ or as $fe = 1 - dfe$ depending on the angle change.

Please note that this option works best with balanced, sinusoidal signals and is available only for three-phase systems.

- *Frequency-locked loop*

  The frequency supplied to the DFT (fe), is estimated based on a double second-order generalized integrator frequency-locked loop (DSOGI-FLL) with gain normalisation as presented in [1]. For the single phase measurement, a SOGI-FLL is used for estimating the frequency.

  The following proportional gain parameters need to be defined:
  - *SOGI proportional gain* (input parameter $Ksogi$)
  - *FLL proportional gain* (input parameter $Gfll$)

  In some cases, the frequency transients when locking the frequency can be very high, and therefore a rate limiter has been included in order to limit the estimated frequency. If the estimated frequency fe has high transients, a corresponding rate limit in Hz/s (parameter $rateLimit$) can be used in order to provide a ”smoother” transient frequency value to the DFT calculation.

  Please note that this option works very well also with unbalanced signals. The frequency tracking does not work properly for very strong distorted signals.

- *Fast fourier transformation*

  The frequency supplied to the DFT is obtained by a Fast Fourier Transform (FFT) calculation (separately for each signal) at the clocked interval.

  The following input parameters need to be defined:
  - *Number of samples per period* (input parameter $samplesPerPeriod$)
  - *Number of samples for FFT window* (input parameter $samplesPerWindow$)

  The resulting sampling frequency in Hz, with which the calculation buffer is filled at defined intervals, is obtained using the nominal frequency $Fnom$ and the input parameter $samplesPerPeriod$ as:

$$samplingFreq = Fnom \cdot samplesPerPeriod \tag{20}$$

  The resulting FFT window length in s of the buffer used in the calculation is obtained using the nominal frequency $Fnom$ as:

$$windowLength = \frac{samplesPerWindow}{samplesPerPeriod \cdot Fnom} \tag{21}$$

  Using a moving FFT on the buffer gives us the frequency of the signal with which the DFT is executed.

  Please note that this method has a limited bandwidth and may produce inaccurate results if the frequency deviates from the nominal frequency by more than $\pm 10\%$.

The outputs are updated at constant intervals defined by the parameter *Calculate outputs every* (input parameter $samplesForUpdate$) which defines after how many samples the DFT is calculated (and outputs are updated). The resulting update cycle (parameter $clockPeriod$) in s is:

$$clockPeriod = \frac{samplesForUpdate}{samplesPerPeriod \cdot Fnom} \tag{22}$$

When the option *Smooth outputs when updated* is used, the outputs are linearly interpolated (smoothed).

The available outputs for the DFT based measurement are presented in Table 4.5.

Table 4.5: Digital outputs, RMS phasor and symmetrical components (phase quantities) for EMT-Simulation

Name	Unit	Description
i_A	p.u.	RMS current A
i_B	p.u.	RMS current B
i_C	p.u.	RMS current C
$i1$	p.u.	Positive sequence current
$i2$	p.u.	Negative sequence current
$i0$	p.u.	Zero sequence current

The available calculation quantities for the DFT based measurement are presented in Table 4.6.

Table 4.6: Calculation quantities, RMS phasor and symmetrical components (phase quantities) for EMT-Simulation

Name	Unit	Description
ir:A	p.u.	RMS current A, real part
ir:B	p.u.	RMS current B, real part
ir:C	p.u.	RMS current C, real part
ii:A	p.u.	RMS current A, imaginary part
ii:B	p.u.	RMS current B, imaginary part
ii:C	p.u.	RMS current C, imaginary part
F:A	Hz	Frequency of the current signal A
F:B	Hz	Frequency of the current signal B
F:C	Hz	Frequency of the current signal C

4.3.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase DFT based measurement are presented in Table 4.7.

Table 4.7: Digital outputs, RMS phasor and symmetrical components (phase quantities) for EMT-Simulation, single-phase measurement (*Number of phases*$= 1$)

Name	Unit	Description
i	p.u.	RMS current

The available calculation quantities for a single-phase DFT based measurement are presented in Table 4.8.

Table 4.8: Calculation quantities, RMS phasor and symmetrical components (phase quantities) for EMT-Simulation, single-phase measurement (*Number of phases*$= 1$)

Name	Unit	Description
ir	p.u.	RMS current, real part
ii	p.u.	RMS current, imaginary part
F	Hz	Frequency of the current signal

4.4 Alpha, beta and gamma components

The alpha, beta and gamma current components are calculated using the Clarke transformation.

The available measured outputs are presented in Table 4.9.

Table 4.9: Outputs, Alpha, beta and gamma components for EMT-Simulation, single terminal

Name	Unit	Description
ir	p.u.	Current, alpha-component
ii	p.u.	Current, beta-component
i	p.u.	Current, magnitude (using the alpha and beta component)
$i0$	p.u.	Current, gamma-component

A symmetrical three-phase current with an RMS-value of $1kA$ (peak value of $\sqrt{2}\ kA$) will result in a current with a magnitude of $\sqrt{ir + ii} = 1p.u.$, assuming a rated current of $1kA$. Hence, the output of RMS- and EMT-simulation is compatible, which is important for models that shall be used in both simulation modes.

The output currents for the EMT simulation are calculated depending on the phase technology of the terminal as follows:

- 3-phase system:
  The phase currents are transformed using the $\alpha\beta\gamma$ transformation:

$$ir = i_\alpha = \frac{1}{3} \cdot (2 \cdot i_a - i_b - i_c) \tag{23}$$

$$ii = i_\beta = \frac{1}{3} \cdot (\sqrt{3} \cdot i_b - \sqrt{3} \cdot i_c) \tag{24}$$

$$i0 = i_\gamma = \frac{1}{3} \cdot (i_a + i_b + i_c) \tag{25}$$

- BI-phase system ($180°$):

$$ir = \frac{1}{2} \cdot (i_a - i_b) \tag{26}$$

$$ii = 0 \tag{27}$$

$$i0 = \frac{1}{2} \cdot (i_a + i_b) \tag{28}$$

- 2-phase system ($120°$):

$$ir = \frac{1}{\sqrt{3}} \cdot (i_a - i_b) \tag{29}$$

$$ii = 0 \tag{30}$$

$$i0 = \frac{1}{\sqrt{3}} \cdot (i_a + i_b) \tag{31}$$

- Single-phase (*Number of phases*$= 3$):

  The current is set (transformation is not used) to the current i available from the EMT-Simulation as follows:

$$ir = iu \tag{32}$$
$$ii = 0 \tag{33}$$
$$i0 = 0 \tag{34}$$

- For DC systems, only one signal is available ir ($ii = 0$, $i0 = 0$).

4.4.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase measurement (*Number of phases*$= 1$) are presented in Table 4.10.

Table 4.10: Outputs, Alpha, beta and gamma components for EMT-Simulation, single-phase measurement

Name	Unit	Description
ir	p.u.	Current
ii	p.u.	Current (set to 0)
i	p.u.	Current, magnitude

- Single-phase measurement (*Number of phases*$= 1$):

  The current is set (transformation is not used) to the current i_{phase} available from the EMT-Simulation for the selected measured phase (*a, b, c, or n*) as follows:

$$ir = i_{phase} \tag{35}$$
$$ii = 0 \tag{36}$$

5 References

[1] R. Teodorescu, M. Liserre, and P. Rodriguez. *Grid Converters for Photovoltaic and Wind Power Systems*. Wiley, 2011.

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